THE CANONICAL CECH-TYPE COMPLEX OF PUSHFORWARDS ATTACHED TO AN AUGMENTATION OF SIMPLICIAL SITES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the adjointness used to identify Hom(G, a_{n,*}F_n) with the Hom against the resolution terms is the standard pullback-pushforward adjunction for the augmentation and its components (invoked at source lines 2008-2009); the resolution complex itself is established on card L46164 (lemma-simplicial-resolution-augmentation); no verification is omitted.

Editorial note. The label lemma-augmentation-cech-complex names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial-site situation and augmentations of sites. Same chapter, Situation \label{situation-simplicial-site} and Remark \label{remark-augmentation-site}. Statement lines 1988-1991: a_0 is an augmentation towards a site D as in the remark, giving the topos morphism a with pushforward a_* and component pushforwards a_{n,*} on the total site C_total.
- The simplicial resolution attached to an augmentation. Same chapter, \label{lemma-simplicial-resolution-augmentation} (card L46164). Statement lines 1928-1943: for any abelian sheaf G on D there is an exact complex ... -> g_{2!}(a_2^{-1}G) -> g_{1!}(a_1^{-1}G) -> g_{0!}(a_0^{-1}G) -> a^{-1}G -> 0 of abelian sheaves on C_total; applied here via Hom(-, F).

Source reference. Original reference 0D78: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1929–1929. Prerequisite pointer invoked at proof line(s) 2011; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D6Z: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1125–1125. Prerequisite pointer invoked at proof line(s) 1991; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1989; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let a_0 be an augmentation towards a site $\mathcal{D}$ as in Remark 0D6Z. For an abelian sheaf $\mathcal{F}$ on $\mathcal{C}_{total}$ there is a canonical complex*

$$0 \to a_*\mathcal{F} \to a_{0,*}\mathcal{F}_0 \to a_{1,*}\mathcal{F}_1 \to a_{2,*}\mathcal{F}_2 \to \ldots$$

on $\mathcal{D}$ which is exact in degrees $-1, 0$ and exact everywhere if $\mathcal{F}$ is injective.

Proof. To construct the complex, by the Yoneda lemma, it suffices for any abelian sheaf $\mathcal{G}$ on $\mathcal{D}$ to construct a complex

$$0 \to \mathrm{Hom}(\mathcal{G}, a_*\mathcal{F}) \to \mathrm{Hom}(\mathcal{G}, a_{0,*}\mathcal{F}_0) \to \mathrm{Hom}(\mathcal{G}, a_{1,*}\mathcal{F}_1) \to \ldots$$

functorially in $\mathcal{G}$. To do this apply $\mathrm{Hom}(-, \mathcal{F})$ to the exact complex of Lemma 0D78 and use adjointness of pullback and pushforward. The exactness properties in degrees $-1, 0$ follow from the construction as $\mathrm{Hom}(-, \mathcal{F})$ is left exact. If $\mathcal{F}$ is an injective abelian sheaf, then the complex is exact because $\mathrm{Hom}(-, \mathcal{F})$ is exact. $\square$